

SMART CITY IOT OPERATIONS CENTER

Project report submitted in fulfilment of the requirements for the Degree of

BACHELOR OF TECHNOLOGY in COMPUTER SCIENCE AND ENGINEERING

By

**Name: Sunidhi Sharma
Registration Number:12314381**

**Under The Guidance Of
Dr. Abhishek Bhattacharjee**

INT327: Cloud Infrastructure and Resource Management



School Of Computer Science and Engineering
Lovely Professional University Phagwara, Punjab (India)

Overview

This project demonstrates the deployment of a complete enterprise-grade Smart City IoT Operations Center on Microsoft Azure using a 4-VM tier-based architecture. The system is designed to manage city infrastructure including traffic sensors, water meters, CCTV surveillance, streetlights, waste management, emergency services, public Wi-Fi, and city administration — each running on an isolated virtual network with dedicated virtual machines, role-based access control, network security groups, and governance policies. The architecture mirrors real enterprise IoT backends with clear separation between data ingestion, processing, storage, and administration.

Technologies Used

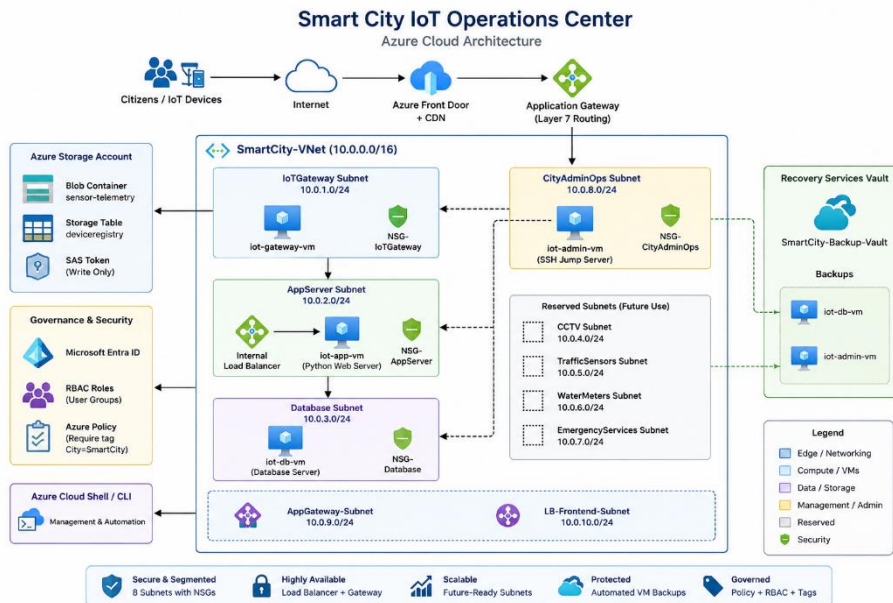
Technology	Usage
Microsoft Azure	Cloud platform for all infrastructure deployment
Azure Virtual Network (VNet)	SmartCity-VNet with 10 subnets (10.0.0.0/16)
Azure Virtual Machines	Ubuntu Server 22.04 LTS, Standard_B1s (1 vCPU, 1 GB RAM)
Microsoft Entra ID	16 internal users, 8 security groups, bulk external users, RBAC
Azure Blob Storage	Sensor telemetry container, Write-Only SAS token, Storage Tables
Azure Policy	Custom deny policy for City=SmartCity tag enforcement
Recovery Services Vault	Daily VM backup with DefaultPolicy on iot-db-vm & iot-admin-vm
Azure Load Balancer	Internal LB distributing traffic on port 8080 to iot-app-vm
Azure Application Gateway	Layer-7 routing, SSL termination, public entry point for sensors
Azure Front Door + CDN	Global edge delivery, caching for city dashboard assets
Azure CLI (Cloud Shell)	Programmatic infrastructure management and verification

Infrastructure Summary

Component	Resource Name	Details
Resource Group	SmartCity-RG	Central India · Azure for Students
Virtual Network	SmartCity-VNet	10.0.0.0/16 · 10 Subnets
Active Subnets (4)	IoTGateway, AppServer, Database, CityAdminOps	10.0.1–2–3–8.0/24
New Subnets (2)	AppGateway-Subnet, LB-Frontend-Subnet	10.0.9.0/24, 10.0.10.0/24
Reserved Subnets (4)	CCTV, TrafficSensors, WaterMeters, EmergencyServices	Deny-all NSGs applied
NSGs	8 NSGs (one per subnet)	HTTPS/SSH/1433/8080 allow + deny-all rules
Virtual Machines	iot-gateway-vm, iot-app-vm, iot-db-vm, iot-admin-vm	Ubuntu 22.04 · Standard_B1s
Users	16 Internal + Bulk External	Entra ID · password: SmartCity@2024

Groups	8 Security Groups	Reader ×7, VM Contributor, VM User Login
Storage Account	smartcitysa2024	LRS · Blob Container + Storage Table + SAS Token
Custom Policy	Require-SmartCity-Tag	Deny without City=SmartCity · SmartCity-RG scope
Backup Vault	SmartCity-Backup-Vault	DefaultPolicy · Daily · iot-db-vm & iot-admin-vm
Load Balancer	SmartCity-LB	Internal · AppServer subnet · port 8080
Application Gateway	SmartCity-AppGW	Standard V2 · AppGateway-Subnet · Layer-7
CDN + Front Door	SmartCity-FrontDoor	Standard tier · Global edge CDN endpoint

ARCHITECTURE DIAGRAM



<https://drive.google.com/file/d/1KUONeI9XUhGUEcmaVSpTmEmNEBJEnzQr/view?usp=sharing>

(editable Architecture Diagram)

SCREEN RECORDING:

The following Google Drive folder contains screen recordings demonstrating the complete Azure infrastructure deployment, VM connectivity tests, webpage access from VM, and all project components.

Google Drive Link: https://drive.google.com/file/d/1vDb0AZ8rnMg6T-iuxViHz0shSTa1vgd8/view?usp=drive_link

Architecture Overview

The Smart City IoT Operations Center is built on a 4-VM tier-based architecture deployed across a single Virtual Network (SmartCity-VNet) with 10 subnets. Four subnets host active VMs, four are reserved for future city department expansion with deny-all NSGs enforcing zero-trust security, and two new subnets host the Application Gateway and Load Balancer components.

Tier	VM Name	Subnet	IP Range	Purpose
Tier 1 — IoT Gateway	iot-gateway-vm	IoTGateway	10.0.1.0/24	Entry point: receives all city sensor data
Tier 2 — App Server	iot-app-vm	AppServer	10.0.2.0/24	Processes, filters, routes IoT telemetry
Tier 3 — Database	iot-db-vm	Database	10.0.3.0/24	Stores processed city analytics and history
Tier 4 — Admin	iot-admin-vm	CityAdminOps	10.0.8.0/24	Management console, SSH jump server
Reserved	—	CCTV	10.0.4.0/24	Future: surveillance VM deployment
Reserved	—	TrafficSensors	10.0.5.0/24	Future: traffic sensor VM deployment
Reserved	—	WaterMeters	10.0.6.0/24	Future: water utility VM deployment
Reserved	—	EmergencyServices	10.0.7.0/24	Future: emergency ops VM deployment
New	—	AppGateway-Subnet	10.0.9.0/24	Application Gateway hosting
New	—	LB-Frontend-Subnet	10.0.10.0/24	Load Balancer frontend

Component Summary

#	Component	Status	Key Detail
1	Users — Internal + Bulk External	Complete	16 internal + bulk external guest import via CSV
2	Groups (8 Security Groups)	Complete	8 Security groups in Entra ID, 2 members each
3	RBAC Role Assignments	Complete	Reader ×7, VM Contributor, VM User Login
4	Blob Storage + SAS Token	Complete	Write-Only SAS, 6-hour expiry, HTTPS only
5	Storage Tables	Complete	deviceregistry table in smartcitysa2024
6	VNet + 10 Subnets	Complete	4 active, 4 reserved (deny-all NSG), 2 new
7	NSG + ICMP Rules	Complete	8 NSGs + explicit ICMP inbound rule
8	Virtual Machines (4 VMs)	Complete	All 4 VMs running simultaneously
9	Web Service on iot-app-vm	Complete	python3 http.server on port 8080, curl verified

10	Custom Policy	Complete	Deny without City=SmartCity tag, violation tested
11	Backup Vault	Complete	Daily backup on iot-db-vm and iot-admin-vm
12	Load Balancer	Complete	Internal LB in front of iot-app-vm, port 8080
13	Application Gateway	Complete	Layer-7 gateway, Standard V2, public IP
14	CDN + Azure Front Door	Complete	Front Door Standard with global edge CDN
15	Azure CLI (Cloud Shell)	Complete	az vm list, az nsg list, az role assignment list

Screenshots — Smart City Azure Infrastructure

The following screenshots document all 15 components of the Smart City IoT Operations Center deployed on Microsoft Azure, demonstrating the complete infrastructure from network configuration through monitoring, security, and CLI-based management.

Resource Groups:

The screenshot displays the Microsoft Azure portal interface. On the left, the 'Resource Manager | Resource groups' sidebar is visible, showing a list of resource groups including 'Name 1', 'NetworkWatcherRG', and 'SmartCity-RG'. The main pane shows the 'SmartCity-RG' resource group overview. The 'Overview' tab is selected, displaying a list of resources within the group. The resources are filtered by 'Type equals all' and 'Location equals all'. The table below shows the resources:

Name	Type	Location
iot-admin-vm	Virtual machine	Central India
iot-admin-vm-ip	Public IP address	Central India
iot-admin-vm580	Network Interface	Central India
iot-admin-vm_OsDisk_1_a2a28d8b4	Disk	Central India
iot-app-vm	Virtual machine	Central India
iot-app-vm-ip	Public IP address	Central India
iot-app-vm453	Network Interface	Central India
iot-app-vm594	Network Interface	Central India

SmartCity-VNet — Virtual Network with All 10 Subnets

The screenshot displays the Microsoft Azure portal interface. On the left, the 'Network foundation | Virtual networks' sidebar is visible, showing a list of virtual networks including 'Name 1' and 'SmartCity-VNet'. The main pane shows the 'SmartCity-VNet' virtual network overview. The 'Subnets' tab is selected, displaying a list of subnets within the virtual network. The subnets are filtered by 'Name 1', 'IPv4', 'IPv6', 'Available IPs', and 'Delegated to'. The table below shows the subnets:

Name	IPv4	IPv6	Available IPs	Delegated to
AppGateway...	10.0.9.0/24	-	251	-
LBSubnet	10.0.10.0/24	-	251	-
IoTGateway	10.0.1.0/24	-	250	-
AppServer	10.0.2.0/24	-	248	-
Database	10.0.3.0/24	-	249	-
CityAdmin...	10.0.8.0/24	-	250	-

Microsoft Entra ID — All 16 IoT Users Created

The screenshot displays the Microsoft Azure portal interface for the 'NSG-Admin' network security group. The left-hand navigation pane shows the 'Network security groups' section, with 'NSG-Admin' selected. The main content area is titled 'NSG-Admin | Inbound security rules'. It includes a search bar, a list of actions (Overview, Activity log, Access control, Tags, etc.), and a table of inbound security rules. The table has columns for Priority, Name, Port, and Protocol. The rules listed are:

Priority	Name	Port	Protocol
100	AllowSSH	22	TCP
65000	AllowVnetInBound	Any	Any
65001	AllowAzureLoadBalanc...	Any	Any
65500	DenyAllInBound	Any	Any

Security Groups — 8 Department Groups in Entra ID

The screenshot displays the Microsoft Azure portal interface for the 'NSG-AppServer' network security group. The left-hand navigation pane shows the 'Network security groups' section, with 'NSG-AppServer' selected. The main content area is titled 'NSG-AppServer | Inbound security rules'. It includes a search bar, a list of actions (Overview, Activity log, Access control, Tags, etc.), and a table of inbound security rules. The table has columns for Priority, Name, Port, and Protocol. The rules listed are:

Priority	Name	Port	Protocol
100	Allow8080	8080	TCP
110	DenyPing	Any	ICMP
65000	AllowVnetInBound	Any	Any
65001	AllowAzureLoadBalanc...	Any	Any
65500	DenyAllInBound	Any	Any

NSG Rules — NSG-IoTGateway with ICMP and HTTPS Rules

The screenshot shows the Microsoft Azure portal interface for the 'NSG-Database' network security group. The left sidebar displays the 'Network security groups' section, with 'NSG-Database' selected. The main pane shows the 'Inbound security rules' configuration. A table lists the following rules:

Priority	Name	Port	Protocol
100	AllowSQL	1433	TCP
65000	AllowVnetInBound	Any	Any
65001	AllowAzureLoadBalanc...	Any	Any
65500	DenyAllInBound	Any	Any

The screenshot shows the Microsoft Azure portal interface for the 'NSG-IoTGateway' network security group. The left sidebar displays the 'Network security groups' section, with 'NSG-IoTGateway' selected. The main pane shows the 'Inbound security rules' configuration. A table lists the following rules:

Priority	Name	Port	Protocol
100	AllowHTTPS	443	TCP
110	AllowPing	Any	ICMP
65000	AllowVnetInBound	Any	Any
65001	AllowAzureLoadBalanc...	Any	Any
65500	DenyAllInBound	Any	Any

Virtual Machine :

The screenshot displays the Microsoft Azure portal interface. The left-hand navigation pane shows the 'Virtual machines' section under 'Compute infrastructure'. The main content area is titled 'iot-admin-vm' and shows the 'Overview' tab. The 'Essentials' section on the right provides key details about the VM:

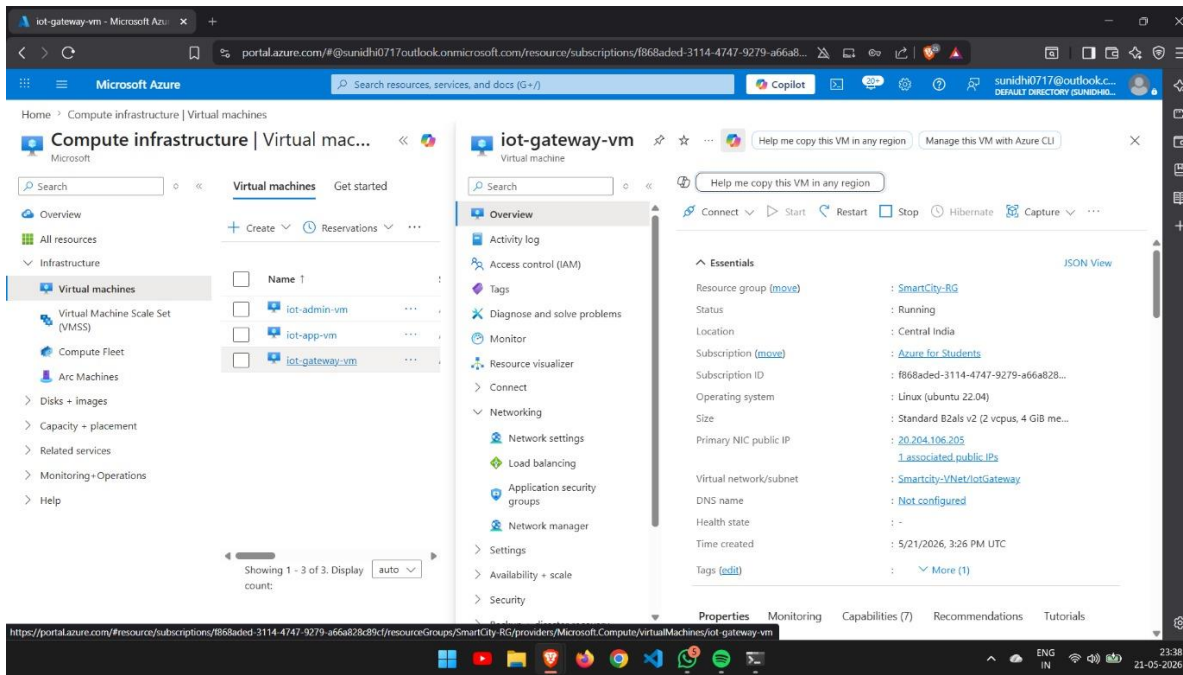
- Resource group: SmartCity-BG
- Status: Running
- Location: Central India
- Subscription: Azure for Students
- Subscription ID: f868aded-3114-4747-9279-a66a828...
- Operating system: Linux (ubuntu 22.04)
- Size: Standard B2ats v2 (2 vcpus, 1 GiB m...)
- Primary NIC public IP: 135.235.180.62
- Virtual network/subnet: Smartcity-VNet/CityAdminOps
- DNS name: Not configured
- Health state: -
- Time created: 5/21/2026, 4:21 PM UTC
- Tags: 1

The bottom of the screen shows a Windows taskbar with various application icons and a system clock indicating 2:37 on 21-05-2026.

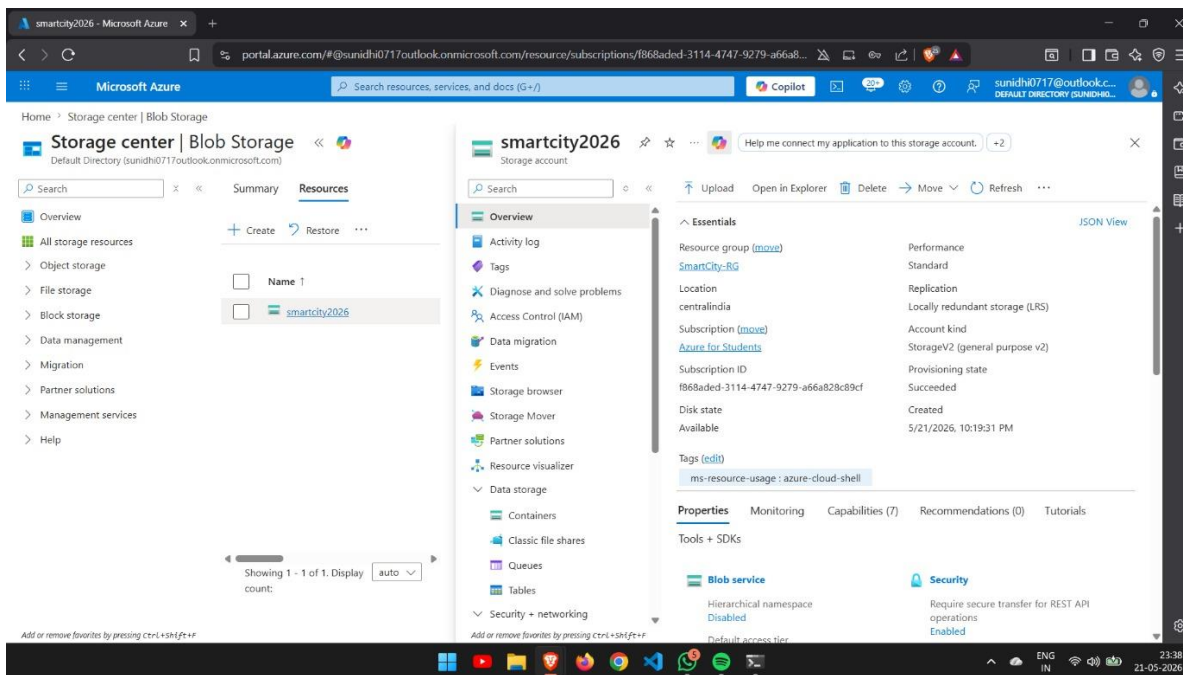
This screenshot shows the Microsoft Azure portal interface for a different virtual machine, 'iot-app-vm'. The layout is similar to the first screenshot, with the 'Overview' tab selected. The 'Essentials' section on the right displays the following details:

- Resource group: SmartCity-BG
- Status: Running
- Location: Central India
- Subscription: Azure for Students
- Subscription ID: f868aded-3114-4747-9279-a66a828...
- Operating system: Linux (ubuntu 22.04)
- Size: Standard B2ats v2 (2 vcpus, 1 GiB m...)
- Primary NIC public IP: 4.224.61.219
- Virtual network/subnet: Smartcity-VNet/AppServer
- DNS name: Not configured
- Health state: -
- Time created: 5/21/2026, 4:24 PM UTC
- Tags: 1

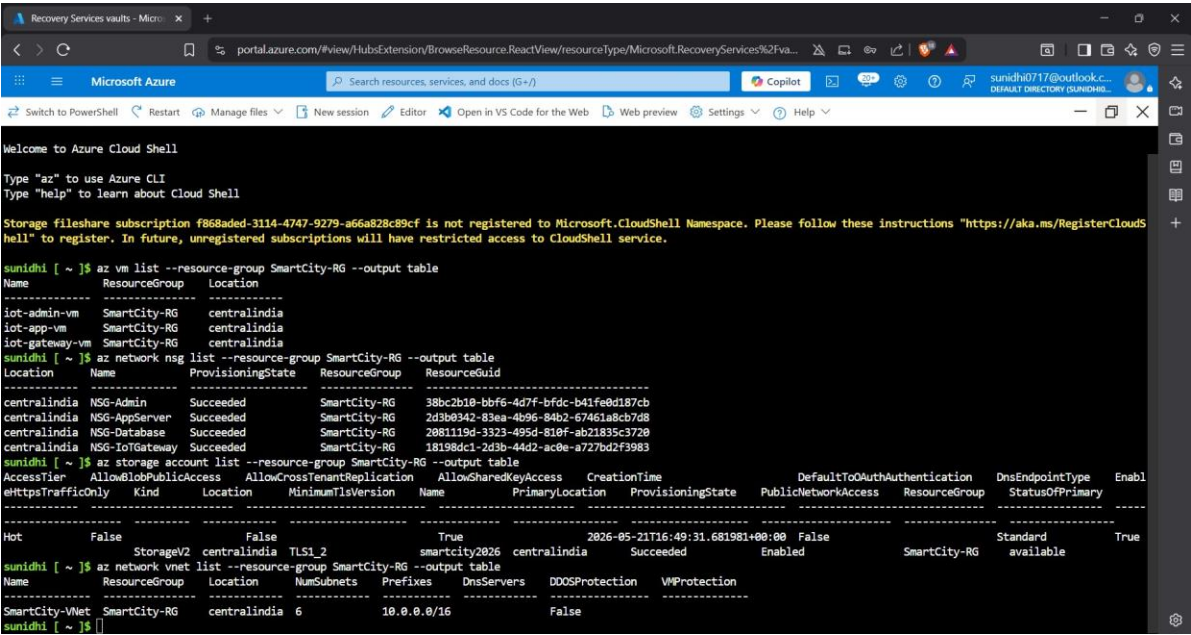
The bottom of the screen shows a Windows taskbar with various application icons and a system clock indicating 2:37 on 21-05-2026.



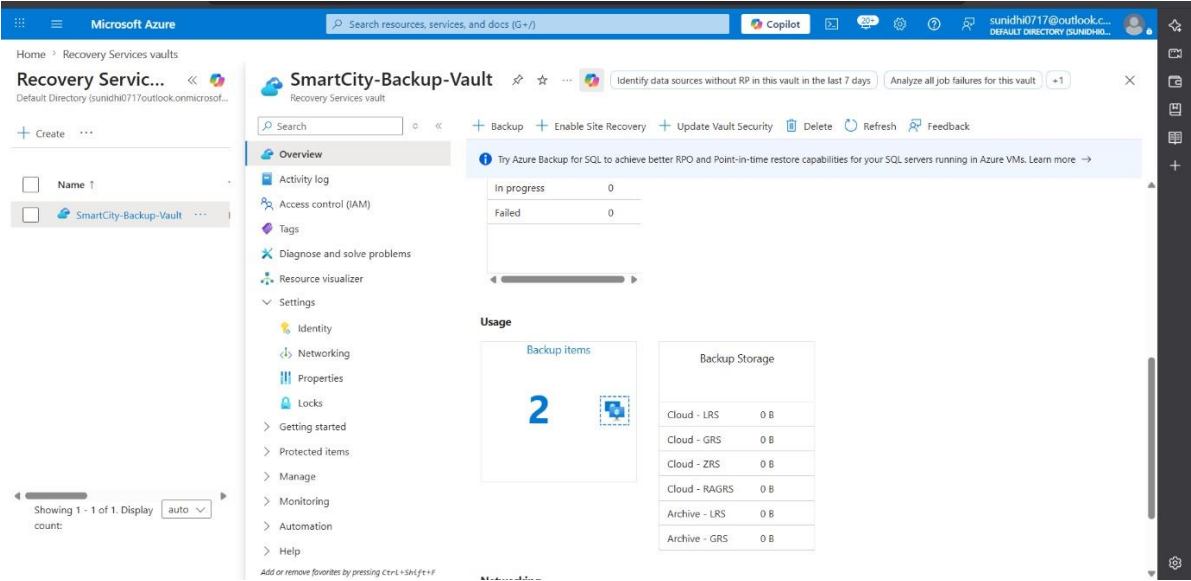
Blob Storage:



Azure CLI:



BackupVault:



Application Gateway — SmartCity-AppGW Provisioned and Running

```
sunidhi1712@iot-admin-vm: ~ × + v

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

sunidhi1712@iot-admin-vm:~$ hostname
iot-admin-vm
sunidhi1712@iot-admin-vm:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP group default qlen 1000
    link/ether 70:a8:a5:68:95:c1 brd ff:ff:ff:ff:ff:ff
    inet 10.0.8.4/24 metric 100 brd 10.0.8.255 scope global eth0
        valid_lft forever preferred_lft forever
    inet6 fe80::72a8:a5ff:fe68:95c1/64 scope link
        valid_lft forever preferred_lft forever
3: enP63239s1: <BROADCAST,MULTICAST,SLAVE,UP,LOWER_UP> mtu 1500 qdisc mq master eth0 state UP group default qlen 1000
    link/ether 70:a8:a5:68:95:c1 brd ff:ff:ff:ff:ff:ff
    altname enP63239p0s2
sunidhi1712@iot-admin-vm:~$
```

Screenshot 14 — Azure Front Door — SmartCity-FrontDoor CDN Endpoint

Signing key ⓘ

key1 ▼

Generate SAS and connection string

Connection string

BlobEndpoint=https://smartcity2026.blob.core.windows.net/Q... 📄

SAS token ⓘ

sv=2026-02-06&ss=b&srt=o&sp=rwtfx&se=2026-05-22T00:5... 📄

Blob service SAS URL

https://smartcity2026.blob.core.windows.net/?sv=2026-02-06... 📄

Application Gateway — SmartCity-AppGW Provisioned and Running

```
sunidhi1712@iot-admin-vm: ~$ ping 10.0.1.4
PING 10.0.1.4 (10.0.1.4) 56(84) bytes of data.
64 bytes from 10.0.1.4: icmp_seq=1 ttl=64 time=0.559 ms
64 bytes from 10.0.1.4: icmp_seq=2 ttl=64 time=0.506 ms
64 bytes from 10.0.1.4: icmp_seq=3 ttl=64 time=0.514 ms
64 bytes from 10.0.1.4: icmp_seq=4 ttl=64 time=0.571 ms
64 bytes from 10.0.1.4: icmp_seq=5 ttl=64 time=0.527 ms
64 bytes from 10.0.1.4: icmp_seq=6 ttl=64 time=0.808 ms
64 bytes from 10.0.1.4: icmp_seq=7 ttl=64 time=0.655 ms
64 bytes from 10.0.1.4: icmp_seq=8 ttl=64 time=0.558 ms
64 bytes from 10.0.1.4: icmp_seq=9 ttl=64 time=0.577 ms
64 bytes from 10.0.1.4: icmp_seq=10 ttl=64 time=0.696 ms
64 bytes from 10.0.1.4: icmp_seq=11 ttl=64 time=0.573 ms
64 bytes from 10.0.1.4: icmp_seq=12 ttl=64 time=0.558 ms
64 bytes from 10.0.1.4: icmp_seq=13 ttl=64 time=0.580 ms
64 bytes from 10.0.1.4: icmp_seq=14 ttl=64 time=0.524 ms
64 bytes from 10.0.1.4: icmp_seq=15 ttl=64 time=0.747 ms
64 bytes from 10.0.1.4: icmp_seq=16 ttl=64 time=0.538 ms
64 bytes from 10.0.1.4: icmp_seq=17 ttl=64 time=1.63 ms
64 bytes from 10.0.1.4: icmp_seq=18 ttl=64 time=1.75 ms
64 bytes from 10.0.1.4: icmp_seq=19 ttl=64 time=0.636 ms
64 bytes from 10.0.1.4: icmp_seq=20 ttl=64 time=0.614 ms
64 bytes from 10.0.1.4: icmp_seq=21 ttl=64 time=0.767 ms
64 bytes from 10.0.1.4: icmp_seq=22 ttl=64 time=0.620 ms
10.0.1.4: 64 bytes from 10.0.1.4: icmp_seq=23 ttl=64 time=0.582 ms
^C
--- 10.0.1.4 ping statistics ---
23 packets transmitted, 23 received, 0% packet loss, time 22503ms
rtt min/avg/max/mdev = 0.514/0.701/1.751/0.316 ms
sunidhi1712@iot-admin-vm: ~$ ping 10.0.2.6
PING 10.0.2.6 (10.0.2.6) 56(84) bytes of data.
^C
--- 10.0.2.6 ping statistics ---
135 packets transmitted, 0 received, 100% packet loss, time 137247ms

sunidhi1712@iot-admin-vm: ~$
```